Integrated communication and location monitoring system for vehicle monitoring via smartphone calls
ScienceDirect ID: S1110016825009329

1. Overview of the Paper (Done by Suad)

The paper presents an integrated vehicle monitoring and location tracking system that relies on smartphone-based communication (missed calls or SMS) combined with embedded GPS and GSM modules installed in the vehicle. The core motivation of the research is to address the high cost, complexity, and continuous data dependency of traditional vehicle tracking and telematics systems.

Conventional vehicle monitoring solutions often rely on continuous internet connectivity, cloud servers, and proprietary hardware, which increases operational cost and limits usability in areas with unreliable data networks. The authors propose an alternative approach that uses on-demand communication triggered by a smartphone call or message, enabling vehicle owners or fleet managers to retrieve real-time location data only when required.

The system is designed to be:

- Cost-effective
- Energy efficient
- Easy to deploy in existing vehicles
- Functional even in low-bandwidth or poor internet connectivity environments

Experimental results from the prototype demonstrate reliable operation, fast response times, and acceptable location accuracy, making the solution practical for real-world applications such as personal vehicle security, fleet monitoring, and logistics management.

2. What the Authors Are Proposing(Done by Suad - 4517)

The authors propose a hybrid embedded–mobile communication framework for vehicle monitoring. The key idea is to replace continuous data streaming with a trigger-based location retrieval mechanism.

2.1 Core Components of the Proposed System

2.1.1 Embedded Vehicle Unit

- Microcontroller-based processing unit
- GPS module for location acquisition
- GSM module for SMS and call detection
- Optional OBD-II interface for vehicle integration

2.1.2 Smartphone Interface

- The user initiates a missed call or SMS to the vehicle's GSM module
- No dedicated mobile application is strictly required, reducing complexity

2.1.3 Communication Workflow

- The vehicle system detects the incoming call or SMS
- GPS coordinates are fetched instantly
- Location information is sent back to the user via SMS, typically as a Google Maps link

The novelty lies in leveraging basic cellular services (calls and SMS) rather than continuous internet-based tracking, which significantly reduces cost and power consumption.

3. How It Works (Operational Workflow) (Done by Annanya-4513 and Jeeah-4518)

The proposed system operates in a simple, trigger-based sequence that uses standard GSM and GPS capabilities without relying on continuous internet connectivity.

3.1 Detection of User Trigger

- The monitoring user places a phone call or sends an SMS to the vehicle's GSM-equipped device

- The system continuously listens for incoming calls/SMS and detects when an authorized caller ID matches a pre-configured whitelist

- Only when the incoming call is from an authorized number does the system proceed

3.2 Authorization Verification

- Before any location retrieval, the system checks the caller's identity against a stored list of allowed users

- If the caller is not authorized, the system does not disclose any location information

3.3 GPS Location Acquisition

- Once the call trigger is accepted, the vehicle's GPS module (e.g., NEO-6M) retrieves current geographical coordinates (latitude and longitude)

- The system also records a timestamp so the location data has temporal context

3.4 Message Preparation

The acquired location data is formatted into a human-readable message, such as:

Vehicle Location: 28.6139° N, 77.2090° E at 14:30:22

Alternatively, the coordinates can be encoded as a Google Maps link that opens directly in map applications.

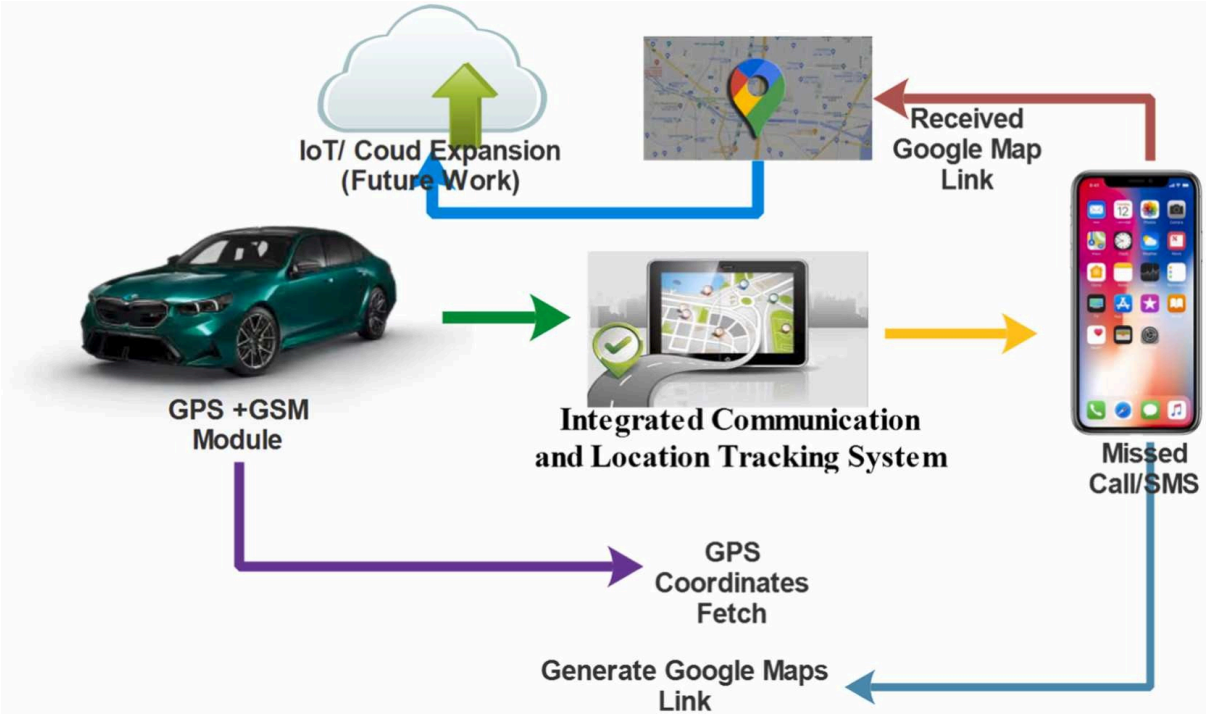
3.5 Transmission of Response

- The formatted location message is sent back through the GSM network to the user as an SMS message

- If GPS acquisition fails, the system attempts a limited number of retries (e.g., up to three). Failure after retries results in a "Location Unavailable" message

3.6 Optional Error Handling

- The system includes mechanisms to handle GPS signal loss (e.g., tunnels or dense urban environments) and GSM communication delays by employing retries or fallback logic



4. Benefits of the Proposed System (Done by Kiara-4514)

4.1 Cost Effectiveness

One of the strongest advantages of the proposed system is its low operational cost. Since it does not rely on continuous mobile data or cloud services, recurring expenses are minimal. SMS-based communication is significantly cheaper than data-based telemetry.

4.2 Reduced Power Consumption

By operating only when triggered by a call or message, the system avoids constant GPS polling and data transmission. This improves energy efficiency and makes it suitable for long-term deployment without frequent maintenance.

4.3 Simplicity and Ease of Deployment

The system can be retrofitted into existing vehicles using standard interfaces like GSM modules and OBD-II ports. The absence of a complex smartphone application simplifies user interaction and reduces compatibility issues.

4.4 Reliability in Low-Connectivity Regions

Unlike internet-dependent systems, SMS and call-based communication works reliably even in regions with weak or unstable data connectivity. This makes the system particularly useful in rural areas and developing regions.

4.5 Fast Response and Acceptable Accuracy

Experimental evaluation reported:

- Response time of less than three seconds
- Location accuracy within approximately five meters

These metrics are sufficient for most vehicle monitoring and security use cases.

5. Challenges and Limitations (Done by Sameera - 4515and Vaidehi-4516)

5.1 Security and Privacy Risks

SMS and GSM-based communication channels are inherently vulnerable to spoofing, interception, and unauthorized access if not properly secured. Without encryption and authentication mechanisms, attackers could potentially request vehicle location data or impersonate authorized users.

5.2 Network Dependency

Although SMS requires less bandwidth than data services, the system still depends on GSM network availability. In areas with no cellular coverage, the system becomes non-functional.

5.3 Limited Scalability

The proposed design works well for individual vehicles or small fleets. However, managing large fleets with frequent location requests could result in:

- Increased SMS traffic
- Network congestion
- Delays in response time

Scaling such a system would require additional backend infrastructure or message-handling mechanisms.

5.4 Functional Limitations

The system focuses primarily on location monitoring. Advanced telematics features such as continuous tracking, driver behavior analysis, route optimization, or predictive maintenance are not directly supported in the current design.

6. Current Implementation and Real-World Relevance (Done by Annanya-4513)

Although the system described in the paper is a prototype, its underlying principles are already reflected in existing technologies.

6.1 Fleet Telematics Systems

Modern fleet management solutions use GPS and cellular networks for real-time tracking. However, most rely on continuous data streaming. The proposed approach offers a lighter alternative for organizations that only need periodic or on-demand location updates.

6.2 Automatic Vehicle Location (AVL) Systems

AVL systems used in public transport and logistics sometimes rely on SMS as a backup communication channel when data networks fail. The paper's approach formalizes this concept into a primary operational mode.

6.3 Vehicle Security and Theft Recovery

On-demand tracking via missed calls or SMS is especially useful in theft recovery scenarios, where immediate location access is more important than continuous tracking.

6.4 Deployment in Developing Regions

In regions where mobile data is expensive or unreliable, SMS-based vehicle monitoring provides a practical and economically viable solution.

7. Summary

The paper proposes a practical, low-cost, and energy-efficient vehicle monitoring system that uses smartphone calls or SMS to trigger real-time location reporting. Its primary contribution lies in reducing dependency on continuous data connectivity while maintaining acceptable accuracy and responsiveness.

While the system offers clear advantages in terms of cost, simplicity, and applicability in low-connectivity environments, it faces challenges related to security, scalability, and feature richness. With further enhancements in security, IoT integration, and intelligent data processing, the proposed approach has strong potential for adoption in personal vehicle security, small-scale fleet management, and smart transportation ecosystems.

Study/Technique	Communication Mode	Location Tracking	Internet Dependency	Special App Required	Limitations
[14]	GSM (SMS)	GPS Tracker	Required	No	High cost, needs server
[15]	SMS	GPS (Smartphone)	No	Yes	SMS cost, app installation needed
[16]	Phone Call (missed call)	Triggered GPS SMS	No	No	Error-prone call handling
[17]	VoIP Call	GPS (Smartphone)	Yes	Yes	Internet needed for VoIP
[18]	IoT Cloud	GPS/On-board Sensors	Yes	Yes	Expensive, complex setup
Proposed System	GSM Call + SMS	GPS (Smartphone)	No	No	Minimal internet reliance, simple, cost-effective